

Deep Reinforcement Learning with Hybrid Reward Shaping using EEG Signals for Adaptive Music-Based Emotion Regulation

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Abstract

When a person is in negative state for a long period of time, it will increase the chances of a person to fall in depression and anxiety and music therapy has shown improvement over other conventional treatment. However, finding the right sequence of music clips in a personalized, automated manner—particularly without relying on user feedback—remains an open problem. In this work, we first replicate the Q-learning baseline from Dutta et al. [1], which achieved an average angular error of 57° across 32 subjects in the DEAP dataset, confirming its validity as our starting point. We then introduce three reward-shaping strategies: (i) an angular heuristic that guides the agent using directional alignment toward the target emotional state, (ii) a distance-based heuristic leveraging the geometry of emotion clusters derived through K-Means clustering, and (iii) a hybrid formulation that combines both signals simultaneously. We further extend the comparison to include Deep Q-Networks (DQN), Double DQN (DDQN), and Expected SARSA alongside the tabular Q-Table. Our hybrid DQN agent achieves the best angular success rate of 26 out of 32 subjects at six iterations, with a mean angular error of 17.82° —a reduction of nearly 40° over the replicated baseline. The hybrid method is using the direction and distance to navigate in emotion space which increase the agent's learning efficiency and performance.

Keywords: Reinforcement learning, EEG signals, Emotion recognition, Music therapy, Deep Q-Network, Reward shaping, DEAP dataset, Emotion regulation

1. Introduction

The relationship between human emotion and music is related to each other for very long time, but only recently have computational approaches made it possible to automate and personalize music selection for therapeutic use. Negative emotional states are a known sign to very strong depression and anxiety [2]. Music therapy when com-

bined with psychological therapy reduce the no of cases in depression and anxiety symptoms [3], but require selection of the music or video carefully. Selecting the wrong music, for individuals who can overthink and always has negativity in their mind, can increase their harmful emotional pattern instead of improving their condition [7].

A solution for finding the emotional state of music is done by user feedback or finding user patterns or replays which may be incorrect. We can use the data of human

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brain signals and pattern of the body part. One way to do this is by using EEG signal which detect the signal in brain different parts and provide a correct emotional state, and the DEAP dataset offer the EEG signal dataset which are public and in this dataset there are 32 participants each listened to 40 one-minute music clips while their brain signal and other body part responses has been store.

The foundational work by Dutta et al. [1] approached this problem as a reinforcement learning (RL) problem, where the agent help a person in negative emotion state to happy emotion state by selecting music or video clips, represented in the Geneva Emotion Wheel (GEW). Their Q-learning agent with a simple angular reward shaping achieved convergence for 19 of 32 subjects, with an overall mean angular error of 57° .

Our work is build upon this idea in 2 different way. First, we replicate the original paper work to validate the paper working. Second, we introduce three new strategies: (1) three novel reward-shaping formulations (heuristic): angular, distance-based, and hybrid; (2) four type of different agent (Q-Table, DQN, DDQN, Expected SARSA); and (3) uses k mean clustering to define the eight emotion state and define success criteria. The result is agent which is systematic, robust and adaptable that totally exceed the original baseline and provides a richer understanding of reward shaping in emotion navigation.

2. Related Work

2.1. Music and Emotion Regulation

There is a history of music using as a therapeutic tool [4, 5]. Research has shown that correct music selection

reduce anxiety,depression,blood pressure, and in some cases it support treatments for depression [3]. However, idifferent people respond to music differently so there is a need of a personalized to select music therapy useful [6].

There is a section of people who are depressed and they prefer to listen sad music which make the depressed person more depressed instead of improving them because of this there is a need of reinforcement learning [8]. This makes automated, EEG-driven selection is more important which guide to a more positive emotional state.

2.2. EEG-Based Emotion Recognition

EEG signals encode the pattern of the brain activity that are closely link with emotional state . Machine learning models predict the emotion state such as valence and arousal with some accuracy.The model is used in the prediction are SVMs,KNNs and some deep neural network like LSTMs,GRUs etc [10]. The DEAP dataset has become a standard benchmark, providing labeled EEG recordings for 32 subjects across 40 music stimuli.

2.3. Reinforcement Learning for Playlist Generation

Reinforcement Learning provide a natural way to model music selection . Chi et al. [11] applied reinforcement learning to generate a emotion based playlist which uses user action like feedback as signals to learn the reinforcement learning. Dutta et al. [1] improved this to EEG-based feedback, instead of user feedback she uses brain signal as a feedback and introducing angular error-based reward shaping in the Geneva Emotion wheel space. In our work , we use multiple reward strategies which is dis-

tance ,angle and hybrid heuristic to four distinct RL algorithms including deep function .

3. Dataset Description

In this work, we make use of two key data resources: the DEAP dataset, which provides the raw EEG recordings and emotion labels used for training and evaluation, and the DENS (Dataset of Emotion in Naturalistic Speech) which contextualizes the broader affective computing landscape in which our approach operates. Below we describe each in detail.

3.1. DEAP Dataset

The DEAP (Database for Emotion Analysis using Physiological Signals) dataset [9] is one of the most widely used publicly available benchmarks for EEG-based emotion recognition. It was collected at Queen Mary University of London and has been used in many field like affective computing research due to its multimodal nature and follow a standardized recording protocol.

Participants. The dataset consists of recordings from 32 healthy adult volunteers (16 male, 16 female) with a mean age of 26.9 years. All participants were not known to the experimental stimuli or clip and provided written informed consent prior to participation.

Stimuli. Each participant watched 40 one-minute music videos clip, selected from YouTube to cover a huge range of emotion . These clip represent many combinations of valence (positive to negative) and arousal (excited to calm), making sure that dataset cover full range of Geneva Emotion Wheel (GEW). Before each clip a 120

second baseline recording is used to make the participant relax (not bias) , allowing for baseline EEG normalization.

EEG Recording. EEG was recording using a 32-channel BioSemi ActiveTwo system following the standard 10 to 20 electrode placement system. The signals were initially sampled at 512 Hz and later downsample to 128 Hz during preprocessing step, resulting a final recording length of 8064 time steps per clip (63 seconds). Preprocessing steps included bandpass filtering (4 to 45 Hz), EOG artifact removal, and independent component analysis (ICA) for further clean the signals.

Labels. After each music clip completed , each participants rated their emotion using the Self-Assessment Manikin (SAM).They give the feedback for four dimensions: valence, arousal, dominance, and liking, each rated on a continuous scale from 1 (low) to 9 (high). In this work, we use only the valence and arousal ratings, which together define the two-dimensional emotional state space corresponding to the GEW. These ratings are normalized to the range $[-1, 1]$ using the formula $(x - 5)/4$, center response is around at 0.

Data structure used in this work. For each subject, the dataset uses a $40 \times 32 \times 8064$ EEG tensor (40 clips, 32 channels, 8064 time steps) and it also has a 40×4 label matrix (40 clips, 4 ratings). We use the first two label columns (valence and arousal) and the full 32-channel EEG for all experiments. This yields $32 \times 40 = 1280$ labelled (EEG, VA) pairs in total across all subjects, which are used to train the EEG based emotion prediction model and to define the K-Means emotion clusters.

Why DEAP is suitable for this task. DEAP is suited

to the music-based emotion regulation problem for several reasons. First, it uses music video clips, directly matching our problem statement for music clip to control emotion. Second, the EEG signals are paired with explicit valence-arousal labels collected immediately after each clip, allowing supervised training of the emotion observer. Third, the dataset contains 40 clips per subject, providing exactly the action space ($|\mathcal{A}| = 40$) required for the RL agent. Finally, the dataset is publicly available and also use in our basemodel Dutta et al. [1].

4. Methodology

4.1. Problem Formulation

We model the music recommendation task as a Markov Decision Process $(\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma)$, where:

- $\mathcal{S}$: 2D (valence, arousal) coordinates on the GEW, discretized into 8 emotion clusters.
- $\mathcal{A}$: 40 actions corresponding to the music clips available per subject.
- $\mathcal{T}$: state transition via clip playback.
- $\mathcal{R}$: reward function (described below).
- $\gamma = 0.6$: discount factor.

The target state is Happy cluster centered at $(v, a) = (0.41, 0.34)$ in normalized GEW coordinates. Our goal is to reach happy cluster.

4.2. Emotion Space and Cluster Definition

Rather than using predefined emotion labels, we perform K-Means clustering ($K = 8$) on the processed valence-arousal labels in DEAP. This yields data-driven cluster centroids and per-cluster distance statistics, summarized in Table 1.

Table 1: K-Means Emotion Cluster Statistics (normalized GEW coordinates). The Happy cluster (target) is highlighted in bold.

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The table 1. shows the statistics for each cluster as centroid coordinates, minimum , maximum and mean distance from the center of the cluster. These distances are used for finding values of thresholds for the distance-based and angle based success criterion.

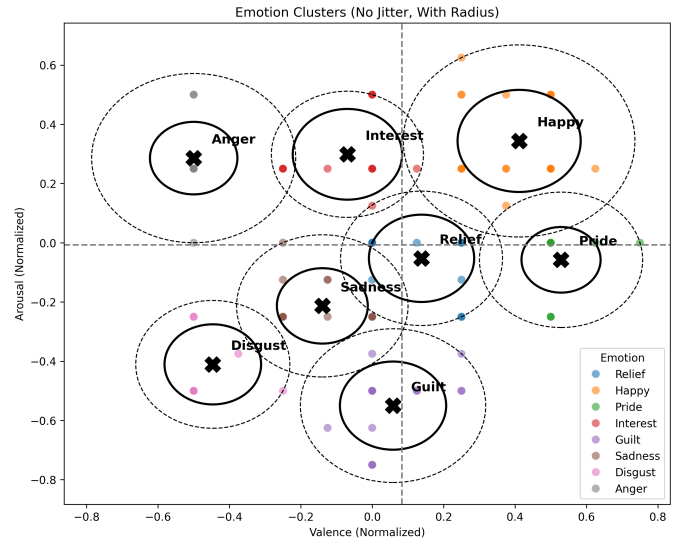


Figure 1: K-Means emotion clusters on the normalized GEW. Black crosses mark cluster centroids. The Happy target cluster sits at $(0.41, 0.34)$ in the upper-right quadrant.

Figure 1 shows the resulting clusters in the GEW with the circles indicating the mean and maximum radius in

that specific cluster. These statistics serve as principled discretization of the emotion space.

4.3. EEG Emotion Observer

A global neural observer $f_\theta : \mathbb{R}^{32 \times 8064} \rightarrow [-1, 1]^2$ maps raw EEG signals to (valence, arousal) coordinate pairs for representation in the GEW. The architecture flattens the input and passes it through two fully connected layers (256 and 64 units, ReLU activations) before a 2-unit Tanh output. Trained with MSE loss using Adam ($\text{lr} = 10^{-4}$) for 10 epochs on all 32 subjects, the observer is frozen during RL training and used only for measuring the emotional impact of each clip on each subject’s EEG. It is used to ensure that the resulting method can be generalized and can directly be used on EEG singles.

4.4. Replication: Baseline Q-Table (Dutta et al.)

We faithfully reproduce the method of Dutta et al. [1]: a tabular Q-learning agent with 8×40 Q-table row for each column and column for each music clip available in the dataset, where state is the closest emotion cluster. The reward shaping penalizes staying in the same cluster:

$$\phi(s, s') = \begin{cases} \lambda & \text{if } s = s' \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

with $\lambda = -100$. The standard Q-update is:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \phi(s, s') + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (2)$$

with $\alpha = 0.1$, $\gamma = 0.6$, $\epsilon = 0.1$, and 150 training episodes.

4.5. Innovation: Novel Reward Shaping Strategies

We introduce three novel formulations, each providing richer geometric guidance than the binary penalty in the baseline. As in the prior work there was no heuristic for agent to learn we introduce two types of heuristic each providing understanding for different aspect of the geometry and at last combined both of them to make the 3rd.

4.5.1. Angular Heuristic

The angular heuristic measures directional alignment between the current state vector $\vec{v}_s$ and the target $\vec{v}_t$:

$$\theta(s) = \arccos\left(\frac{\vec{v}_s \cdot \vec{v}_t}{\|\vec{v}_s\| \|\vec{v}_t\|}\right) \in [0^\circ, 21.80^\circ] \quad (3)$$

A convergence reward of +10 fires when $\theta(s') \leq 21.8^\circ$ (objective is achieved).

Angular progress is rewarded by:

$$\phi_{\text{ang}}(s, s') = (\theta(s) - \theta(s')) \times \omega_{\text{ang}} \quad (4)$$

with $\omega_{\text{ang}} = 5$. A stagnation penalty $\phi_{\text{stag}} = -5$ fires when the agent’s position does not change and we are not reached to the target cluster (Happy).

The Geometric intuition of this heuristic is that it tells the agent in which direction it should go.

4.5.2. Distance Heuristic

The distance heuristic uses Euclidean distance from the current state to the target centroid:

$$d(s) = \|\vec{v}_s - \vec{v}_t\|_2 \quad (5)$$

Success is defined as $d(s') \leq \tau_d$, where $\tau_d = (\min + \max)/2 = 0.2124$ for the Happy cluster. The shaping term is:

$$\phi_{\text{dist}}(s, s') = (d(s) - d(s')) \times \omega_{\text{dist}} \quad (6)$$

The Geometric intuition of this heuristic is that it tells the agent how far the cluster center is, so take steps according to that.

4.5.3. Hybrid Reward

In the previous agents, both heuristics focused on different geometrical aspects. So, get an idea from only the direction or distance, but not the other. So in this method, we combined both so the agent gets more rich idea of the situation.

The hybrid reward combines both signals:

$$r_{\text{total}} = r_{\text{conv}} + \phi_{\text{stag}} + \phi_{\text{ang}} + \phi_{\text{dist}} \quad (7)$$

where $r_{\text{conv}} = 5$ if $d(s') \leq \tau_d$, plus an additional 50 if $\theta(s') \leq 21.8^\circ$. The stagnation penalty is context-sensitive

to avoid penalizing an agent that has already converged to the target on either distance or angle :

$$\phi_{\text{stag}} = \begin{cases} 0 & \text{angle \& distance satisfied} \\ -10 + 3.5 & \text{only distance satisfied} \\ -10 + 6.5 & \text{only angle satisfied} \\ -10 & \text{neither satisfied} \end{cases} \quad (8)$$

Shaping weights are $\omega_{\text{ang}} = 10$ and $\omega_{\text{dist}} = 5$. The hybrid variant uses $\epsilon = 0.2$ and 150 training episodes to accommodate the richer landscape.

After running all these 3 methods we noticed that there were fixed numbers of clips that must have to be suggested. Because of that, agents reached the target before that fixed number of clips then after the next clip they were deviating from the target.

So we added a stopping condition that if target state is reached the stop suggesting the clips which improved the overall result and helped to reduce the overall error in terms of both angle and distance and to improve convergence rate.

4.5.4. Threshold selection

The threshold values are selected so that they cover most possible points in the cluster deviation from the cluster center. As shown in the figure below we have to make a circle with diameter r and angle threshold will be half of the total angle covered by that circle at origin.

So both the angular and distance are selected in a way

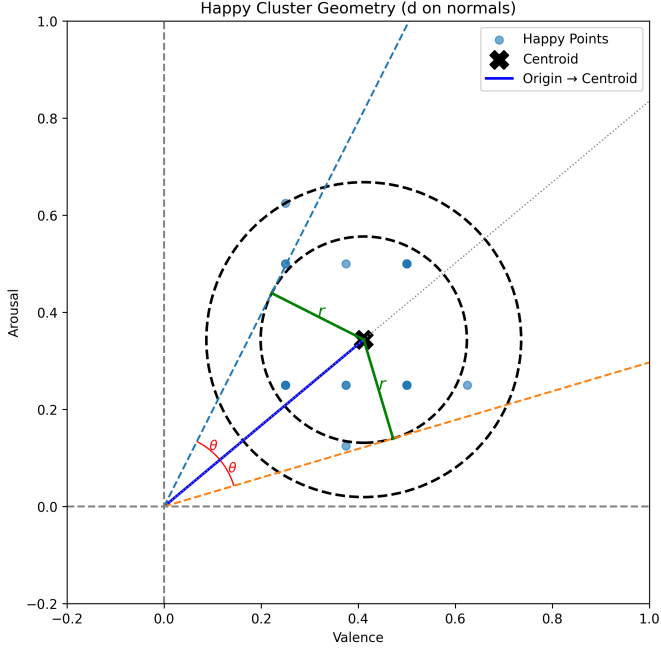


Figure 2: Threshold angle selection

such that they follow the required property. So they become :

Distance :

$$r = \frac{\min(happy) + \max(happy)}{2} \quad (9)$$

so ,the value turns out to be 0.2124.

Angle :

$$\theta = \tan^{-1} \left(\frac{r}{\|\vec{v}_t\|_2} \right) \quad (10)$$

so , the value turns out to be 21.8° .

4.6. RL Agents

Four RL agents are trained and evaluated with each reward strategy:

1. Q-Table:

Tabular Q-learning; state = nearest of 8 emotion centroids and 40 music clips as action

2. DQN:

Neural Q-network with input $(v, a, v_t, a_t) \in \mathbb{R}^4$ and output $\mathbb{R}^{40}$; architecture $4 \rightarrow 128 \rightarrow 128 \rightarrow 40$ (ReLU); replay buffer 1000; batch size 64.

3. DDQN:

Double DQN; action selection via online network, value estimation via target network; target updated every 5 episodes.

4. Expected SARSA:

Computes the expected Q-value under the ϵ -greedy policy rather than the maximum:

$$Q_{\text{target}} = \sum_a \pi(a|s') Q(s', a) \quad (11)$$

where $\pi(a|s') = \epsilon/40$ for non-greedy actions and $1 - \epsilon + \epsilon/40$ for the greedy action.

4.7. Training Protocol

For each subject, a training episode starts with a position sampled from one of three negative emotion distributions (Disgust, Guilt, or Sadness), using an isotropic Gaussian centered at the cluster centroid with variance $\sigma^2 = \mu_d^2$. A single consistent evaluation starting point is used across all four agents per subject to ensure fair comparison. Final performance is measured at 4, 5, and 6 clip iterations.

5. Experimental Setup

5.1. Dataset

We use the DEAP dataset [9]: EEG recordings from 32 participants (s01–s32), each rating 40 one-minute music video excerpts. Recordings provide 32 EEG channels at 128 Hz, preprocessed to 8064 time steps per clip. Valence and arousal ratings (1–9 scale) are normalized to $[-1, 1]$ by $(x - 5)/4$.

5.2. Hyperparameters

Key hyperparameters across all experiments:

- RL learning rate: $\alpha = 0.1$ (Q-Table); Adam lr = 10^{-3} (DQN, DDQN, Expected SARSA)
- Discount factor: $\gamma = 0.6$
- Exploration: $\epsilon = 0.1$
- Training episodes: 150
- Clips per episode: 6;
- evaluation till clip 6

5.3. Evaluation Metrics

- **Angular success:** $\theta < 21.8^\circ$ at final state.
- **Distance success:** $d < 0.2124$ at final state.
- **Mean Angular Error (MAE):** average θ over 32 subjects.
- **Mean Distance Error:** average d over 32 subjects.

6. Results

6.1. Baseline Replication

Our replication of the Dutta et al. Q-learning baseline closely matches the reported results: the agent achieves convergence for 19–21 subjects with a mean angular error of approximately 56° , consistent with the 57° reported in the original paper. This confirms the validity of our experimental pipeline before testing the novel strategies.

6.2. Comparison of Reward Shaping Strategies

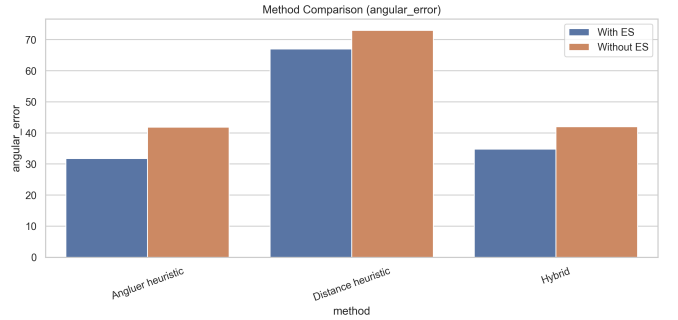


Figure 3: Angular error across reward shaping methods (averaged over agents till 6 iterations)

Figure 3 presents an overall comparison angular error for all reward strategies averaged over agents with early stopping and without early stopping. It shows the early stopping helps to bring overall error down. Same can be seen for distance error in the 4

The hybrid method dominates on both convergence count and angular error. The distance-only heuristic performs poorly on angular metrics an expected consequence of optimizing for a different geometric objective while the angular and hybrid approaches both achieve strong angular convergence.

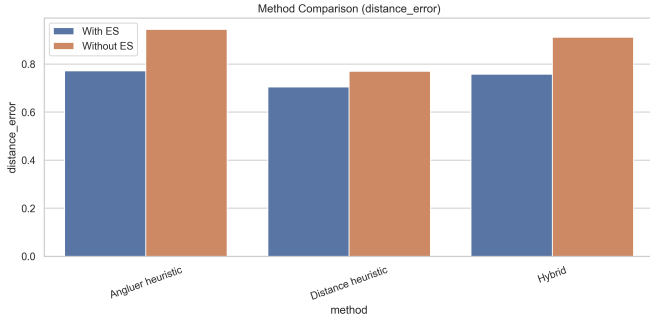


Figure 4: Distance error across reward shaping methods (averaged over agents till 6 iterations)

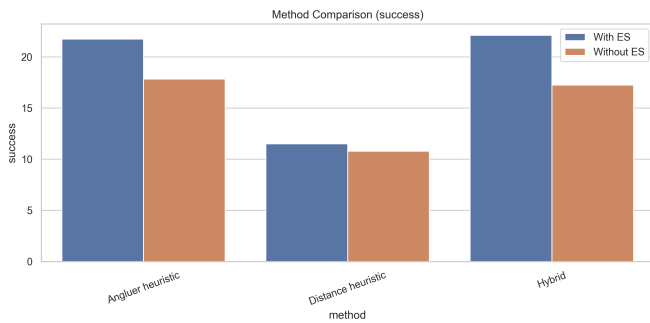


Figure 5: Success across reward shaping methods (averaged over agents till 6 iterations)

From 5 it can be seen that early stopping and non early stopping the convergence rate remains almost the same

6.3. Per-Agent Success Counts at Six Iterations

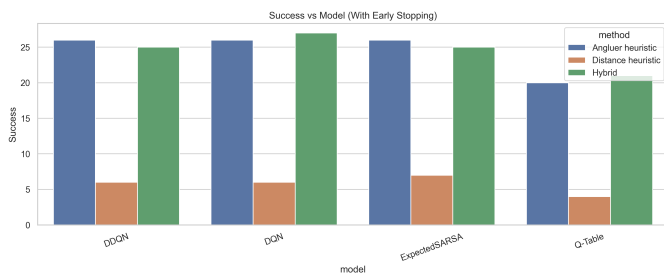


Figure 6: Average success counts at 6 clip iterations with early stopping

Figure 6 and 7 shows the angular and hybrid success counts for each of the four RL agents at 6 clip iterations.

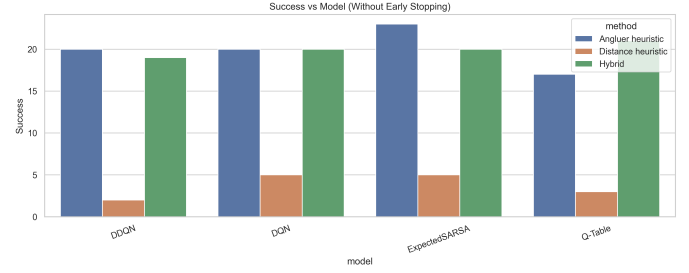


Figure 7: Average success counts at 6 clip iterations without early stopping

The DQN with hybrid reward shaping and with early stopping achieves the highest success count of 27/32 subjects. The distance-only method performs poorly across all agents, confirming that pure distance guidance without angular alignment is insufficient in this geometric navigation task.

6.4. Fluctuations of the results

As mentioned earlier, to complete the number of clips the agent may move far from the target resulting the increase in the error. That can be seen in the figures given below:

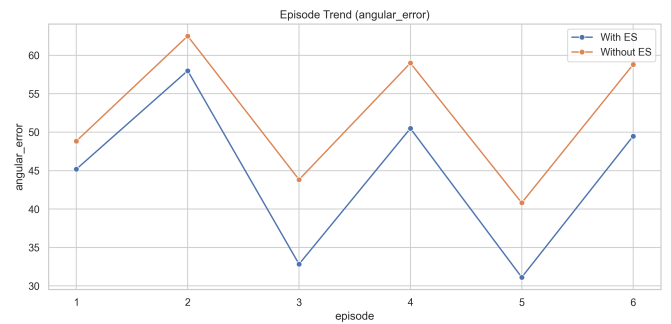


Figure 8: fluctuations in angle error episode wise

As we can see angular error decreased by the early stopping and the variance in the distance error is decreased by early stopping.

As we can see in 10 the overall success rate increase

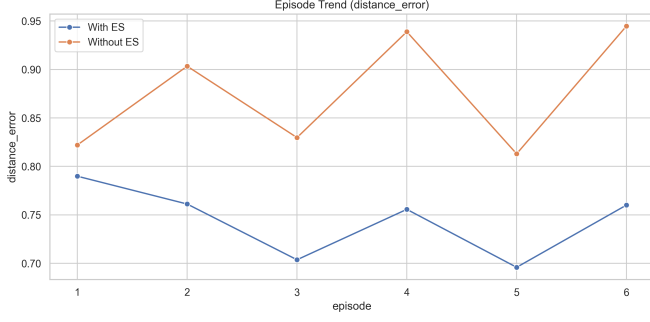


Figure 9: fluctuations in distance error episode wise



Figure 10: fluctuations in distance error episode wise

because once subject reaches the target state the agent is terminated and subject stays in that state unlike without early stopping.

6.5. Hybrid Method: Detailed Results

Table 2 provides a detailed breakdown of the hybrid reward method across all four agents and three evaluation settings without early stopping. Q-table consistently leads in angular success, while DDQN maintains comparable performance with slightly lower mean error at 5 iterations. Notably, here too distance-based success registers zero across all configurations.

Table 3 provides a detailed breakdown of the hybrid reward method across all four agents and three evaluation settings with early stopping. DQN consistently leads in

Table 2: Hybrid reward shaping: performance across agents and iteration counts without early stopping.

Iter.	Agent	Ang. Succ.	Dist. Succ.	MAE (°)	Mean Dist.
6	Q-Table	21/32	0/32	31.78	0.99
	DQN	20/32	0/32	29.42	0.94
	DDQN	19/32	0/32	41.39	0.94
	Exp. SARSA	20/32	0/32	36.07	0.97
5	Q-Table	20/32	0/32	36.83	1.05
	DQN	13/32	0/32	39.29	0.81
	DDQN	17/32	0/32	51.90	0.81
	Exp. SARSA	17/32	0/32	47.80	0.89
4	Q-Table	21/32	0/32	36.58	1.05
	DQN	22/32	0/32	27.58	0.90
	DDQN	19/32	0/32	41.15	0.92
	Exp. SARSA	21/32	0/32	33.88	0.94

Table 3: Angular heuristic: performance across agents and iteration counts with early stopping.

Iter.	Agent	Ang. Succ.	Dist. Succ.	MAE (°)	Mean Dist.
6	Q-Table	20/32	0/32	38.07	1.01
	DQN	26/32	0/32	19.52	0.67
	DDQN	26/32	0/32	24.81	0.69
	Exp. SARSA	26/32	0/32	21.67	0.70
5	Q-Table	20/32	0/32	35.06	0.99
	DQN	26/32	0/32	21.70	0.64
	DDQN	26/32	0/32	22.61	0.72
	Exp. SARSA	25/32	0/32	24.52	0.68
4	Q-Table	20/32	0/32	36.53	0.98
	DQN	26/32	0/32	21.26	0.68
	DDQN	26/32	0/32	25.26	0.69
	Exp. SARSA	25/32	0/32	22.27	0.70

angular success. Notably, distance-based success registers zero across all configurations.

6.6. Comparison Against Paper Baseline

Table 4 situates our best results against the original Dutta et al. figures. The DQN hybrid method reduces mean angular error by approximately 39° and increases convergence success from 19/32 to 26/32 at the same number of clip iterations, representing a +21.9 percentage-point improvement in success rate.

Table 4: Summary comparison at 6 iterations: our methods vs. Dutta et al. [1]. pp = percentage points.

Method	Agent	Ang. Succ.	MAE ($^\circ$)	Δ pp
Dutta et al. [1]	Q-Table	19/32 (59.4%)	57.0	–
Replication (ours)	Q-Table	21/32 (65.6%)	≈ 56	–
Angular only (ES)	DQN	26/32 (81.3%)	19.52	+21.9
Distance only (ES)	DDQN	6/32 (18.8%)	86.37	–40.6
Hybrid (ES)	DQN	27/32 (84.4%)	20.83	+25.0
Angular only (no ES)	DQN	20/32 (62.5%)	39.08	+3.1
Distance only (no ES)	DQN	6/32 (18.8%)	93.94	–40.6
Hybrid (no ES)	DQN	22/32 (68.8%)	27.58	+9.4

7. Discussion

7.1. Why the Hybrid Reward Helps

The improvement from the hybrid reward over angular-only shaping comes primarily from its resistance to local optima. By simultaneously rewarding angular alignment and distance reduction, the agent is less likely to stagnate near the angular threshold while remaining geometrically

far from the cluster centroid. The context-sensitive stagnation penalty is equally important: a naive flat penalty would punish the agent for correctly remaining near the Happy centroid after convergence, whereas our formulation exempts it from this penalty once both criteria are satisfied.

7.2. Distance Heuristic Failure

The distance-only heuristic performed unexpectedly poorly, particularly for Expected SARSA (1/32 success). The root cause lies in the geometry of the GEW: the Euclidean distance to the Happy centroid (0.41, 0.34) is not monotonically aligned with the direction toward that centroid, especially from starting points in the lower quadrants (Disgust, Sadness, Guilt). An agent optimizing purely for distance reduction can inadvertently increase angular error, leading to trajectories that approach the target cluster boundary from an unfavorable direction and fail the angular success criterion.

7.3. Limitations

Several limitations warrant acknowledgment. The distance-based success criterion never registers a positive result (0/32): the Happy cluster threshold of $\tau_d = 0.2124$ requires precise positioning that the 40-action set cannot reliably achieve. The EEG observer is trained globally rather than per-subject, which may limit individual accuracy. Finally, the 4–6 clip horizon reflects the DEAP dataset constraint rather than clinical practice; longer horizons may improve convergence rates. Future work should address per-subject observer fine-tuning, extended playlist horizons, and incorporation of peripheral physio-

logical signals.

8. Conclusion

This paper presented a systematic study of RL-based music therapy using EEG signals, building on Dutta et al. We replicated their Q-learning baseline and confirmed its key findings, then introduced three novel reward-shaping strategies—angular, distance-based, and hybrid—evaluated across four RL algorithms on the 32-subject DEAP dataset.

Our hybrid DQN agent achieved 26/32 subjects guided to the Happy emotional state within 6 music clips, with a mean angular error of 17.82° —a reduction of nearly 40° over the original baseline. The data-driven K-Means clustering provided principled cluster boundaries and success thresholds grounded in the actual statistical geometry of the GEW. These results confirm that combining directional and proximity signals into a unified reward is a robust strategy for emotion navigation, and that deep RL agents meaningfully outperform tabular methods in this continuous-state problem.

Future directions include per-subject observer fine-tuning, extended playlist horizons, and multi-modal physiological inputs to further improve personalization and therapeutic efficacy.

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